

CSI247: Lab 4 – Sorting and Recursion (7 - 9 am)

At the end of this lab you should be able to:

- Creating a utility class
- Implement basic array utility methods
- Follow proper java project folder structure and compile from the command line.
- Create and use arrays
- Implement a sorting method
- Use sorting methods

Lab Question

In this lab, you will implement a utility class with some methods that can be used to sort and analyse some arrays. Use the notes

ArrayUtils

- Create an class called *ArrayUtils* in a package called *lab.utils*.
- Implement a static generic improved bubble sort
- Implement a generic *selection* sort.
- Implement a static generic *binarySum* method.
- Make sure that the generic formal parameter has a *compareTo* method.
- If you have time, implement the static generic *quick* sort methods from the notes in this class.

Tester Class

- Create a class called *Tester* in the *lab* package.
- Create the main method.
- Create 3 arrays of doubles to hold 1000 values.
- Use a for loop and the *Math.random* method to generate floating point values between 0 and 57 and put them into the arrays.
- Use the sorting methods from *ArrayUtils* to sort the arrays and print one of the arrays.